THE m_3–$B^{(2)}$ OBSTRUCTION CLASS: CY-A$_3$ VIA COSTELLO TCFT

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Abstract. The CY-A$_3$ obstruction class vanishes on the ∞-categorical derived centre. The chain-level anticommutators $\{b_k, B^{(2)}\}$ have nonzero individual terms. Costello's TCFT cyclic invariance gives the total identity

$$\sum_k \{b_k, B^{(2)}\} = 0.$$

After formality transport, the chain obstruction represents a Q-exact class in the rational E_3 target. The resulting theorem is an obstruction-class statement, not a termwise chain identity.

Primary theorems

- CY-A$_3$: $\mathrm{Obs}_{A_\infty} = 0$ on the inf-cat derived centre: .
- Individual $\{b_k, B^{(2)}\} \neq 0$ (chain-level): (excludes the per-k identification).
- $\sum_k \{b_k, B^{(2)}\} = 0$ (total sum) via Costello TCFT cyclic invariance: .
- Chain-level m_3 need not be a derivation after formality: (chain-level vs rational scope).

Chapter inputs (pointers)

```
\input{../chapters/theory/m3_b2_obstruction.tex}
\input{../chapters/theory/cyclic_ainf.tex}
\input{../chapters/theory/hochschild_calculus.tex}
\input{../chapters/theory/drinfeld_center.tex}
```

Cross-volume dependencies

- Vol I Theorem $\mathcal{A}^{\infty,2}$ properad (`thm:A-infinity-2`) – provides the Francis–Gaitsgory ambient in which inf-categorical CY-A$_3$ is stated.
- Vol II `thm:chiral-higher-deligne` – the E_3 target operad into which CY-A$_3$ maps.

Structural distinctions

The proof separates three objects: the chain-level anticommutators, the cyclic total sum, and the rational E_3 obstruction class after formality transport. Each arrow in the comparison is verified separately.

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